

EMILY DEEB

Architect, Urban Designer & Computational Analyst | MSc Urban Spatial Science, CASA UCL
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SUMMARY

Architect and urban designer with five years of professional practice at city scale, combining strategic design leadership with geo-computational analysis. Led teams across 15 concurrent projects for municipal and private clients, developing GIS workflows, market analysis tools, and spatial data pipelines that improved office efficiency by 30%. Now advancing into Urban Spatial Science at CASA UCL, with an industry dissertation applying Python-based machine learning to national retail spending datasets in collaboration with CACI Ltd. Brings a rare combination of design intelligence, spatial analytical rigour, and the ability to communicate complex findings to non-technical stakeholders.

EXPERIENCE

Derman Verbakel Architecture – Team Leader – Urban Design & GIS – Tel Aviv, Israel

2020 – 2025

- Led a team of 4 architects across 15 concurrent projects spanning block to city scale, covering strategic masterplans, TOD, urban renewal, housing schemes, and public space for municipal and private clients.
- Built geo-computational workflows in QGIS and ArcGIS for morphological mapping, site feasibility analysis, and profitability modelling.
- Collected, cleaned, and aggregated unstructured spatial datasets for Sakhnin city centre regeneration; produced evidence-based impact assessment reports for the Israeli Authority for Urban Renewal using GIS-based site analysis.
- Trained team members in GIS and data tools; authored strategic planning documentation and managed statutory planning submissions.
- Key projects: Transportation Masterplan of Eilat (Eilat Municipality); Regeneration of Historical City Centre of Sakhnin (1,200 units); TOD Regeneration Plan, Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).

SYL Studio Yigal Levi Architecture – Architect Intern – Jerusalem, Israel

2019 – 2020

- Gained foundational experience in architectural design, documentation, and site coordination.

Buro Happold – Consultancy Project – Building a Better Future Scheme, UCL Bartlett – London, UK

Jun 2026

- Selected through a competitive UCL application to participate in a two-week industry consultancy project on smart campus digital strategy, working directly with Buro Happold's Digital Advisory team.
- Delivered spatial analysis and digital strategy recommendations to the client across a multidisciplinary team; outputs shared with Buro Happold's full UK Advisory team.

CACI Ltd – Dissertation Placement – GeoDS Programme, CASA UCL – London, UK

May 2026 – Aug 2026

- Industry-sponsored dissertation examining whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition.
- Built a Python-based machine learning pipeline using GeoPandas, scikit-learn, SHAP, and spatial autocorrelation to model retail spending patterns at national scale; applied inferential statistics to validate predictive accuracy across CACI proprietary datasets.

Bezalel Academy of Arts and Design – Professor Assistant – Jerusalem, Israel

2023 – 2025

- Co-taught second-year B.Arch design studio, mentoring twenty students through design critique, spatial analysis workshops, and individual guidance.

SKILLS

Spatial Analysis: QGIS, ArcGIS, GeoPandas, Spatial Network Analysis, Spatial Data Science, Google Earth Engine API

Programming: Python (Pandas, GeoPandas, NumPy, Matplotlib), R (RStudio), SQL, React

Machine Learning: Scikit-learn, Classification, Clustering, Regression, SHAP, Pre-modelling Diagnostics

Design & Vis: AutoCAD, Rhinoceros + V-Ray, Adobe Suite, Microsoft Presentation, Data Visualisation, Quarto

Urban Design: Morphometric Analysis, Network Analysis, Block/Plot Typology, Parametric Workflows

Other: Git, Jupyter Notebooks, UI/UX (CalArts certified)

EDUCATION

CASA, University College London

MSc Urban Spatial Science, 2025 – 2026

- Specialising in computational urban analysis, spatial data science, geocomputation, and machine learning for cities.
- Core Modules: GIS & Science | Quantitative Methods | Urban Systems Theory | Dissertation.
- Optional modules: Data Science for Spatial Systems | Building Spatial Apps with Big Data | Urban Data Visualisation | Urban Simulation.

Bezalel Academy of Arts and Design, Jerusalem

B.Arch Architecture, 2015 – 2020

- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation.
- Exchange: Infrastructure Studio, Universität der Künste Berlin (2018). Summer Studio: XL Travelogues, Tokyo (2017).

AWARDS & RECOGNITION

David Azrieli Architecture Prize — Nominated 2020

Bezalel Academy of Arts and Design

Excellence Award — Academic Achievement 2018

Bezalel Academy of Arts and Design

LANGUAGES & CERTIFICATIONS

Arabic (native) | English (C1 — fluent) | Hebrew (fluent)

Licensed Architect — Israeli Labour Department, 2025 | UI/UX Design — CalArts certified